

Vishesh Yadav

Machine Learning Engineer & Data Scientist

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Seoni, Madhya Pradesh 480661

Technical Skills

- **Languages:** C++, Java, Python
- **ML Libraries:** Scikit-learn, Keras, TensorFlow, Pandas, NumPy, Matplotlib, PyTorch
- **Analytics:** Higher Mathematics, Numerical & Complex Analysis, Discrete Mathematics
- **Tools:** IBM Cognos, IBM Watson

Projects

CoreRec & VishGraphs [GitHub](#)

APR 2024

- Designed and deployed a sophisticated graph-based recommendation system serving 500+ daily users, achieving 85% prediction accuracy through innovative neural network architectures and advanced feature engineering
- Spearheaded the development of VishGraphs visualization tool, reducing analysis time by 40% for complex 3D graph rendering while implementing real-time interaction capabilities and custom rendering algorithms
- Implemented cutting-edge graph algorithms utilizing parallel processing techniques, accelerating data processing by 30% while elevating analytical accuracy by 15% through optimized memory management and distributed computing
- Scaled system infrastructure to efficiently process 1M+ nodes and 5M+ edges, driving adoption by 5 research teams across 3 universities, resulting in 10+ published research papers.

Stack Overflow Salary Prediction [Hosted](#)

DEC 2024

- Analyzed and processed 80,000+ Stack Overflow survey responses using advanced NLP techniques, achieving 92% data cleaning accuracy through automated preprocessing pipelines and custom validation algorithms
- Developed a sophisticated regression model incorporating 50+ features, delivering 87% prediction accuracy across 25 countries while accounting for regional economic factors and industry-specific variations
- Built and optimized a Streamlit-based interface handling 1,000+ daily queries with 99.9% uptime, implementing caching mechanisms and load balancing for enhanced performance.
- Generated comprehensive insights from 150+ user feedback sessions, implementing 12 major feature improvements including real-time currency conversion and industry-specific salary benchmarking

VISH - Syntax Handler [Hosted](#)

NOV 2023

- Created an advanced Transformer-based architecture executing 1000+ queries/second with 95% accuracy, incorporating multi-head attention mechanisms and position-wise feed-forward networks
- Optimized self-attention mechanisms through innovative matrix operations, reducing context analysis time by 60% while maintaining model interpretability across diverse syntax patterns
- Enhanced sequence processing speed by 3x through sophisticated parallel computation innovations, including custom CUDA kernels and distributed training pipelines.
- Improved syntax handling accuracy by 40% through implementation of novel tokenization strategies and attention mechanisms, validated across 10,000+ diverse test cases.

Leukemia Detection CNN [GitHub](#)

SEP 2023

- Constructed a state-of-the-art CNN architecture achieving 94% accuracy in leukemia detection across 10,000+ medical images, incorporating residual connections and attention mechanisms
- Integrated and fine-tuned VGG and ResNet architectures with custom layers to reduce false positives by 35%, while maintaining high sensitivity for early-stage detection
- Developed an efficient multi-threaded preprocessing pipeline handling 200+ images/minute with 99% accuracy, implementing advanced augmentation techniques and normalization strategies
- Decreased diagnosis time by 75% while maintaining 96% accuracy across 3 healthcare partnerships, implementing automated reporting and integration with hospital management systems

Contributions

axlearn Repository [Apple](#)

Open Source Contributor NOV 2023

- Enhanced test suite performance by 40% through implementation of comprehensive validation strategies, parallel test execution, and automated regression testing frameworks
- Implemented robust error handling mechanisms with detailed logging and monitoring, reducing system crashes by 65% while improving debugging capabilities
- Optimized core algorithms and data structures to increase model inference speed by 25%, while reducing memory footprint through efficient tensor operations

Certificates

- IBM Building a Chatbot (OCT 2023) - [Certificate](#)
- 2nd runner up in GD conducted by Thought Theater (OCT 2023)

Education

Master in Computer Application (AI&ML) [LPU](#)

Jalandhar, Punjab 2023 – Present

BSc Mathematics [D.P Chaturvedi College](#)

Seoni, MP 2019 – 2022

CGPA: 8.8

Diploma in Computer Applications [SECT Computers](#)

Seoni, MP 2019 – 2021

CGPA: 7.5

12th Science [S.T Francis of Assisi School](#)

Seoni, MP 2018 – 2019

CGPA: 6.3